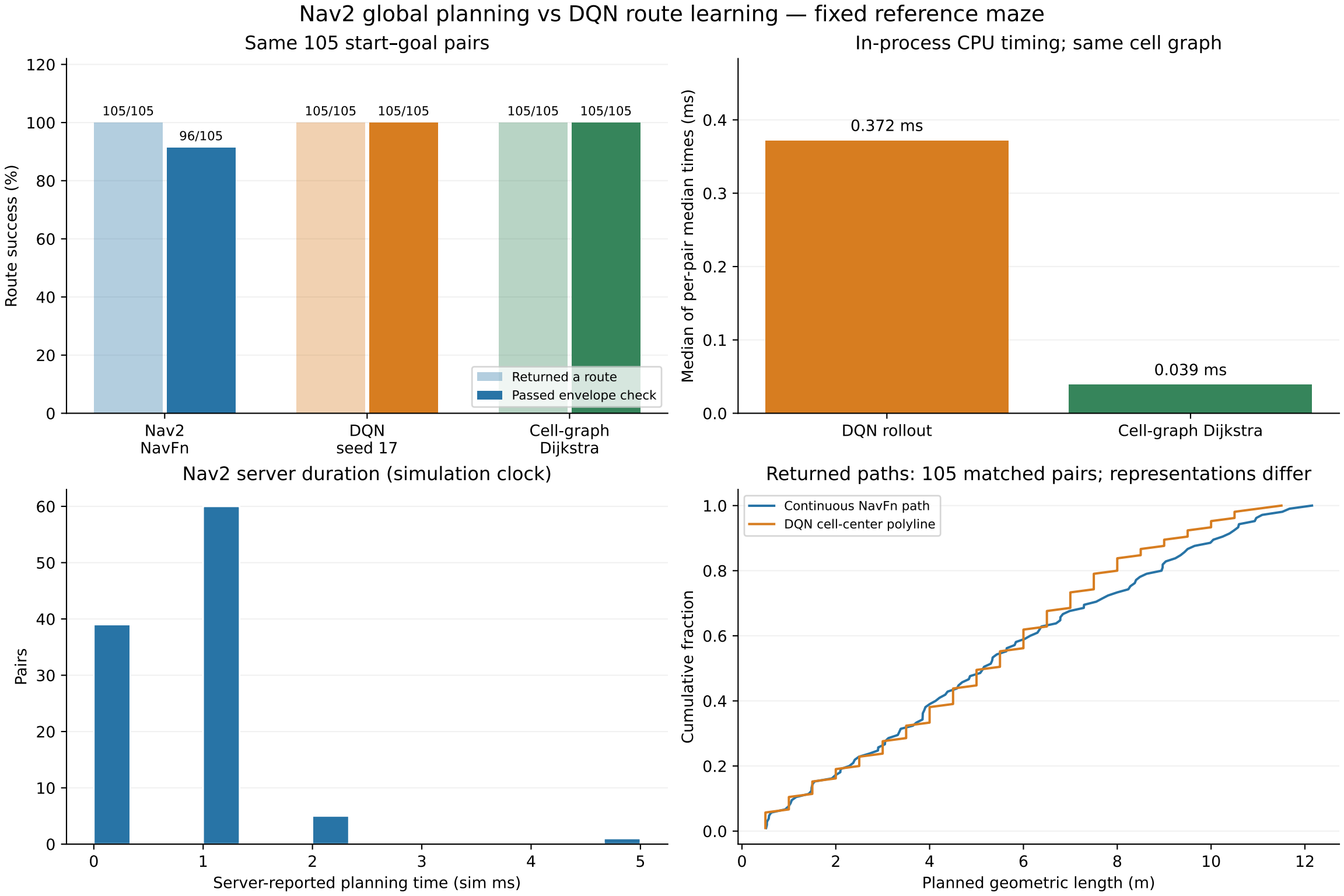
Nav2 global planning vs DQN route learning — fixed reference maze
Same 105 start–goal pairs
Route success (%)
105/105
96/105
105/105
105/105
105/105
105/105
Returned a route
Passed envelope check
Nav2 NavFn
DQN seed 17
Cell-graph Dijkstra
In-process CPU timing; same cell graph
Median of per-pair median times (ms)
0.372 ms
0.039 ms
DQN rollout
Cell-graph Dijkstra
Nav2 server duration (simulation clock)
Pairs
Server-reported planning time (sim ms)
Returned paths: 105 matched pairs; representations differ
Continuous NavFn path
DQN cell-center polyline
Cumulative fraction
Planned geometric length (m)